

MR-BSP Results Discussion: Experiments 25–29

Research discussion copy

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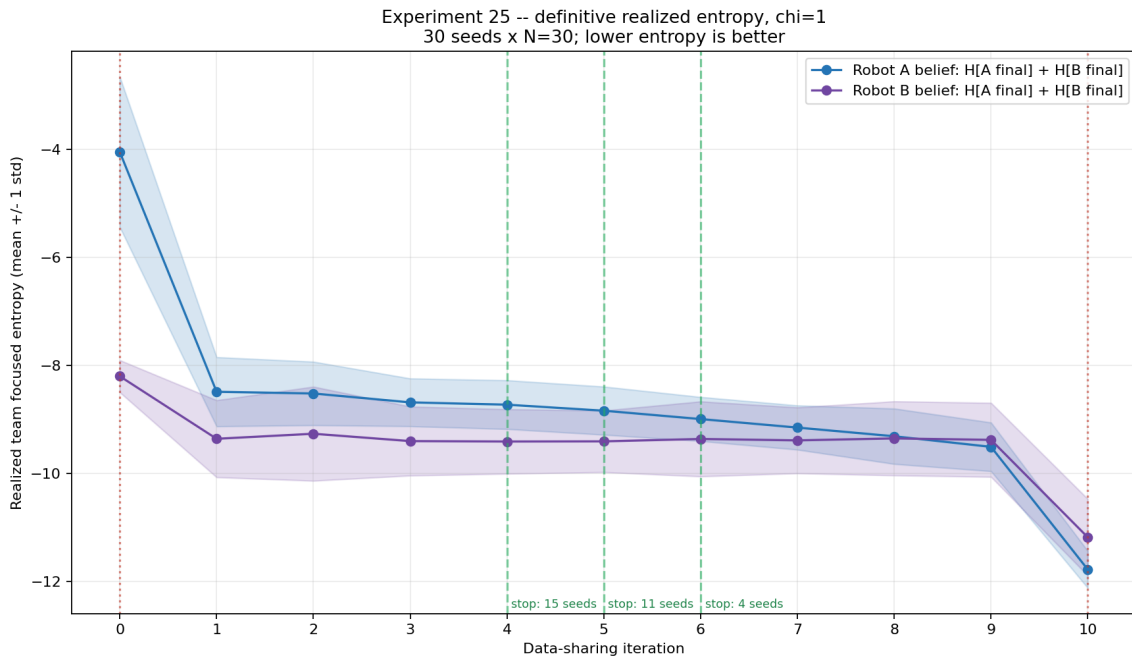
Experiment map

Exp.	Protocol	What it shows
24	Single-session historical	Original dashboard result; its Issue-11-corrected duplicate was archived when slot 27 was repaired.
25	Single-session, 30 seeds, $\chi = 1$	Local execution landmarks included in realized entropy.
26	Single-session, 30 seeds, $\chi = 0$	Known-landmarks-only ablation.
27	Sequential replanning, seed 0	Baseline-I, Baseline-II and R-VerifyAC ($\epsilon = 0.7$); two belief samples. The three trajectory plots below belong here.
28	Sequential replanning pilot, seed 0	Two baselines and $\epsilon = 0.3, 0.6, 0.9$; three belief samples in the completed local files.
29	Modular paired-seed matrix	Proposed comprehensive run; not a completed multi-seed result.

Experiment 25: single planning session

Mean $\pm$ SD over 30 seeds. Each entropy is the sum of the two final-pose focused entropies from one named robot’s local execution belief. Lower is better; negative differential entropy is valid.

Policy	Sharing iterations	Robot A entropy	Robot B entropy
Baseline-II	1.000 ± 0	-4.042 ± 1.393	-8.199 ± 0.269
R-VerifyAC	5.633 ± 0.706	-8.788 ± 0.442	-9.420 ± 0.467
Baseline-I	11.000 ± 0	-11.781 ± 0.319	-11.184 ± 0.488



Experiment 27: sequential replanning, seed 0

The comparison table reports terminal realized robot-local entropy. In the R-VerifyAC session trace below, parentheses instead give planning-time projected focused team entropy H for the executed action in the session’s final logged Step-1 evaluation. A frozen action may differ from that evaluation’s preferred action. Sessions are zero-based; new rounds exclude the initial decision evaluation. This is a single-seed example with two belief samples, not a distributional result. Policies can follow different trajectories and finish in different numbers of physical sessions.

Policy	Sessions	Inconsistent	A final entropy	B final entropy
Baseline-I	25	0	−11.020	−15.074
R-VerifyAC, $\epsilon = 0.7$	19	9	−8.662	−9.121
Baseline-II	26	14	−5.631	−6.255

Session	New rounds	A action (Step-1 H)	B action (Step-1 H)	Agree?
0	9	2 (−14.609)	0 (−14.624)	No
1	1	3 (−14.640)	3 (−14.640)	Yes
2	1	1 (−14.666)	1 (−15.068)	Yes
3	0	0 (−14.535)	0 (−14.677)	Yes
4	1	3 (−9.159)	3 (−14.604)	Yes
5	10	2 (−9.211)	0 (−14.580)	No
6	1	1 (−10.014)	0 (−14.573)	No
7	1	0 (−9.915)	0 (−14.122)	Yes
8	10	2 (−9.783)	3 (−14.038)	No
9	10	0 (−10.702)	3 (−14.111)	No
10	10	0 (−12.852)	1 (−14.383)	No
11	10	0 (−13.089)	2 (−14.156)	No
12	0	1 (−13.137)	1 (−14.591)	Yes
13	10	3 (−12.674)	0 (−13.723)	No
14	1	1 (−13.485)	0 (−14.100)	No
15	0	0 (−13.511)	0 (−13.758)	Yes
16	0	0 (−13.467)	0 (−13.581)	Yes
17	0	0 (−13.512)	0 (−13.638)	Yes
18	0	0 (−13.592)	0 (−13.677)	Yes

The diagrams are all from Robot A’s perspective. Dashed paths are prior history and solid paths are executed continuation. The final uncertainty rings are 2σ isotropic 3D volume-equivalent markers inferred from focused entropy, *not* XY covariance ellipses. Green/red session markers denote consistent/inconsistent decisions. In this seed, R-VerifyAC lies between baselines for both robot-local entropies. Its B-to-A sharing frontier reaches B33 and its plot audit counts 122 fewer B-to-A projection factors than Baseline-I (28% fewer).

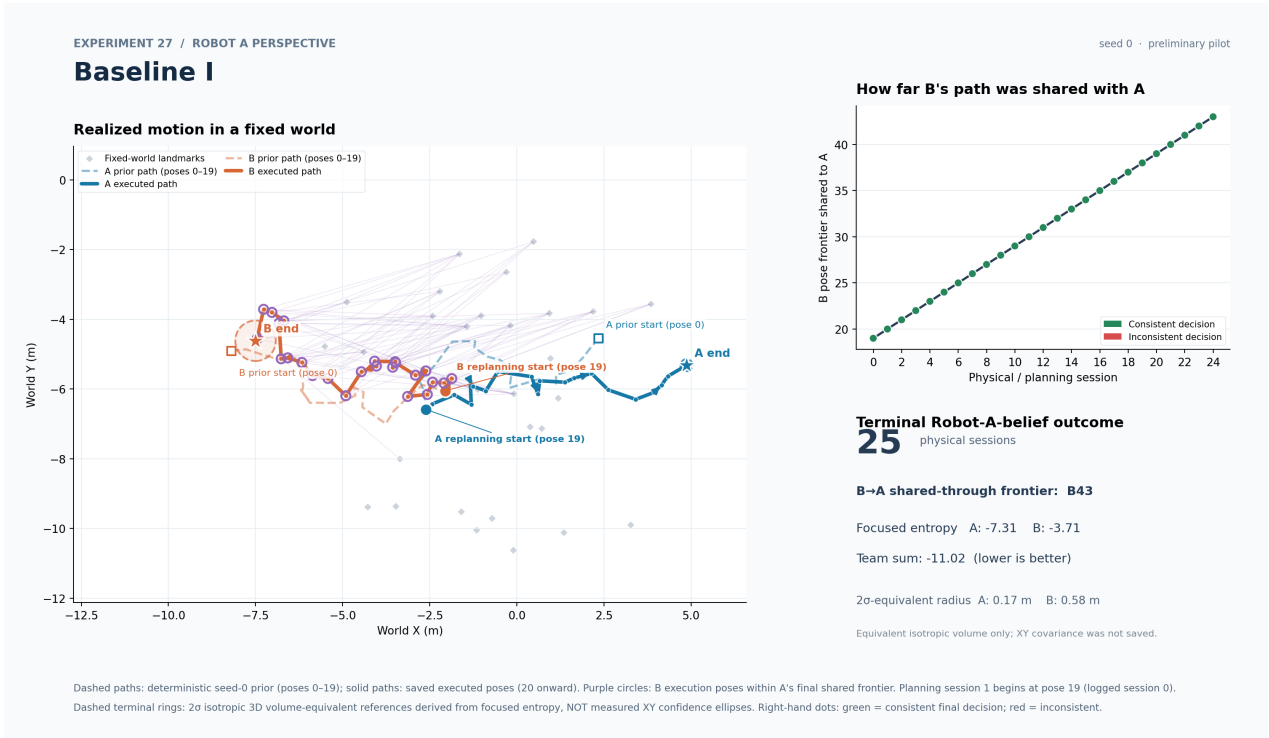


Figure 1: Experiment 27 Baseline-I (full sharing).

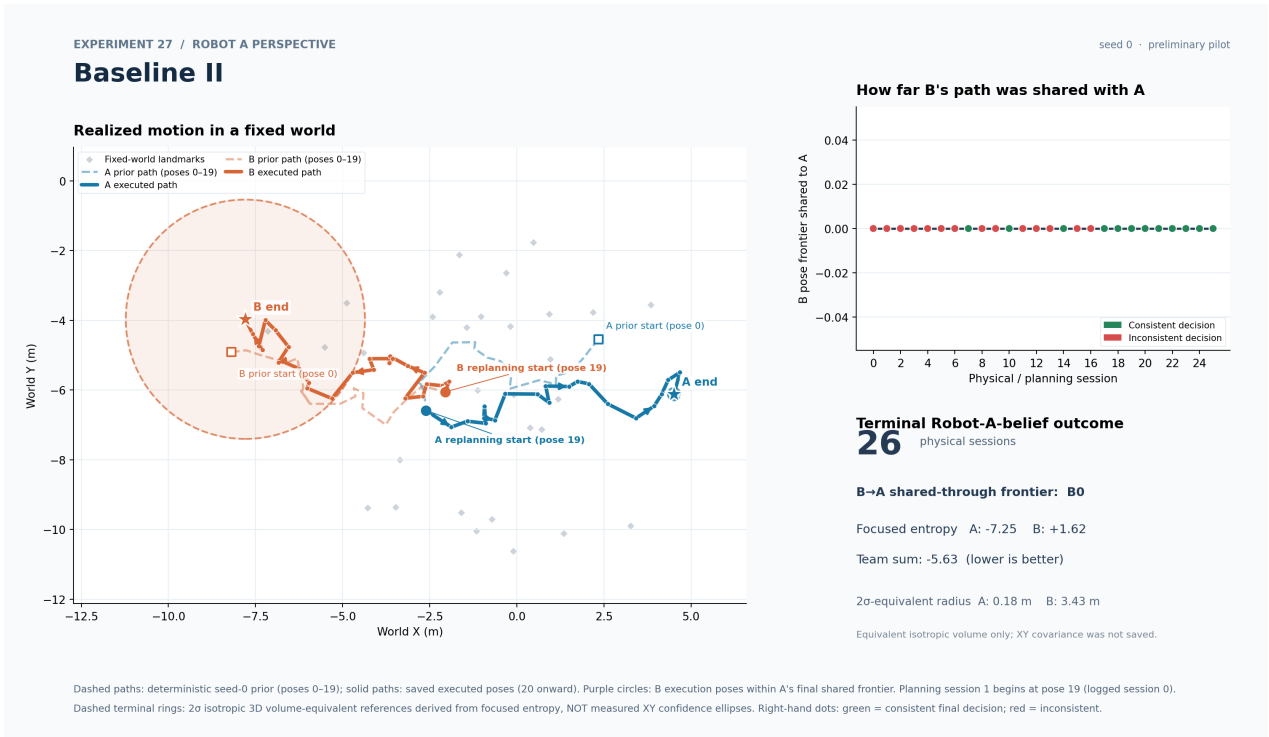


Figure 2: Experiment 27 Baseline-II (no post-initial sharing).

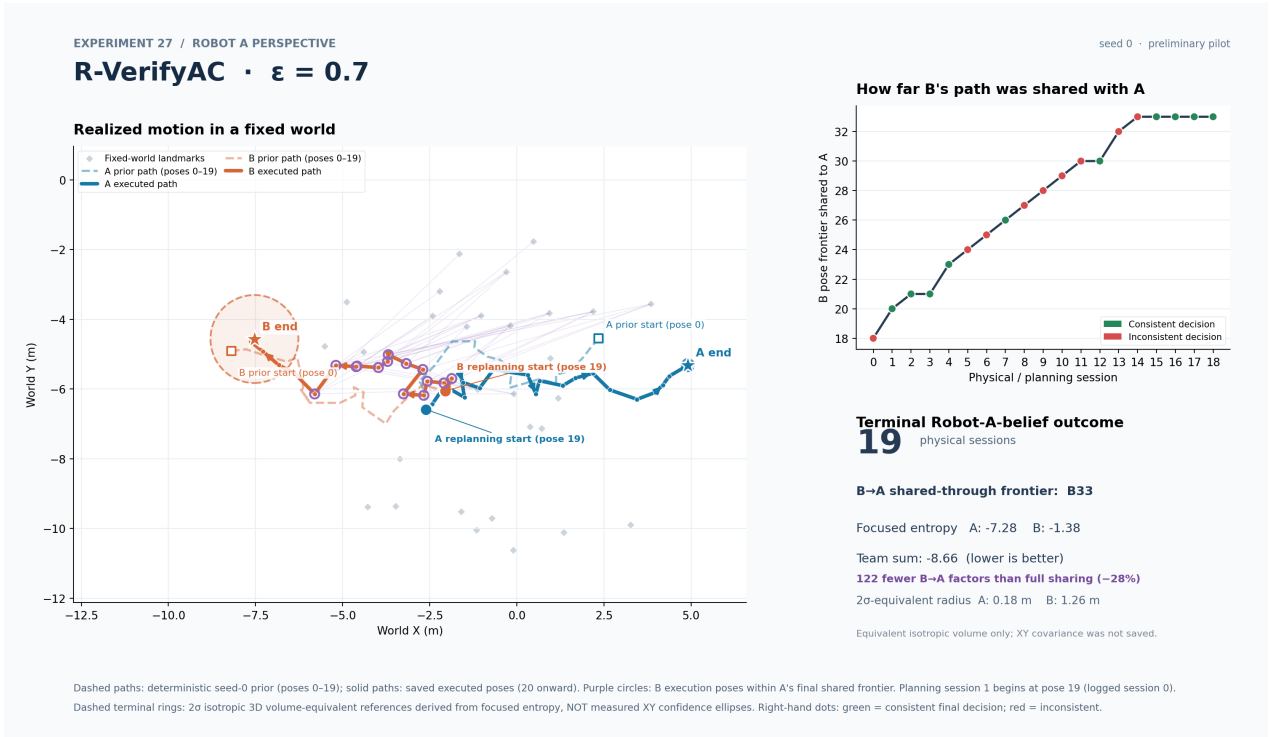


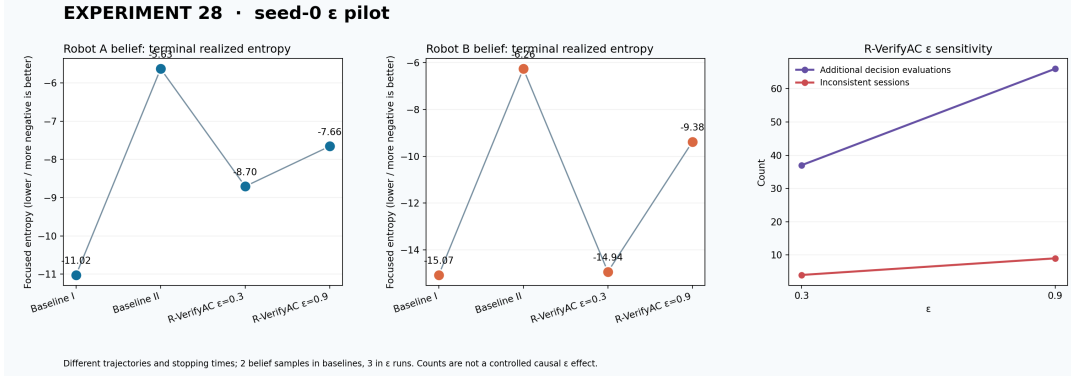
Figure 3: Experiment 27 R-VerifyAC, $\epsilon = 0.7$.

Experiment 28: epsilon pilot, seed 0

The five completed local files use three belief samples. The previously published dashboard comparison figure has only $\epsilon = 0.3$ and 0.9 and uses two-sample baseline files; do not mistake it for a like-for-like five-cell figure.

Policy	Sessions	Inconsistent	Extra evals	A realized H	B realized H
Baseline-I	25	0	—	−11.020	−15.074
R-VerifyAC, $\epsilon = 0.3$	19	4	37	−8.705	−14.936
R-VerifyAC, $\epsilon = 0.6$	19	4	37	−8.705	−14.936
R-VerifyAC, $\epsilon = 0.9$	26	9	66	−7.657	−9.380
Baseline-II	26	14	—	−5.631	−6.255

Extra evals count R-VerifyAC decision evaluations after the initial evaluation in each session, not transmitted factors. The baselines do not run this verification loop, so a dash is more precise than zero comparable evaluations.



Interpretation and reproducibility

Experiment 25's 30-seed single-session means must not be compared as though they were the same estimator as Experiment 27/28's seed-0 sequential terminal outcomes. The epsilon pilot does not establish a smooth entropy–communication trade-off; $\epsilon = 0.3$ and 0.6 have identical reported terminal metrics in this seed. Multiple paired seeds are needed.

Experiment 27 raw result files are in ANPLOS `mr_results/landmark_analysis`: the seed-0 Baseline-I, Baseline-II, and `terminalfixed_durable` JSON outputs dated 2026-09-16. Experiment 28 uses the corresponding seed-0, `samples3` JSON files dated 2026-09-17. The Experiment 28 recorded source state is commit `7d7c9b0` plus dirty/untracked source hashes, so the commit alone is insufficient for bit-for-bit reproduction. Experiment 29 should run independent (seed,policy) cells at a fixed belief-sample count and preserve per-cell timing, progress, and results.